

Superfan Identification & Business Impact Analysis

A data-driven examination of Spotify Reserved:
scoring superfans, modelling retention impact,
and building the financial case for scale.



ANALYST

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\$484MIncremental ARR
(BaseCase, Full Scale)**45%**Reserved Offer
Acceptance Rate**+6%**90-Day Premium
Retention Uplift**0.87**LR Model
AUC-ROC

1.1 Project Overview

On May 21, 2026, Spotify announced **Reserved** at its annual Investor Day - a feature that uses streaming behaviour data to identify an artist's most dedicated fans and hold two concert tickets for each of them, granting a private purchase window before the general public sale. This project builds a transparent analytical framework: scoring superfans from public signals, stress-testing the model at A-list scale, simulating the 90-day Premium retention impact via A/B test framework, and modelling the financial case for global rollout.

Key findings:

- **Score distribution is strongly right-skewed** - the top 5% per artist are clearly separable, validating the p95 eligibility threshold.
- **Repeat plays and playlist saves dominate** - these two signals account for 55% of the composite score and are the most reliably measurable.
- **Developing-artist fans score higher per listener** - smaller fanbases with more intense devotion make Reserved most impactful for emerging acts.
- **A-list scale creates 1,000:1 competition ratios** - three fairness mechanisms are proposed and analysed.
- **Retention uplift is statistically significant** - A/B simulation shows $p < 0.05$ at +6% 90-day retention for offer recipients.
- **Financial case is robust across all scenarios** - even the bear case (3%) produces \$242M incremental ARR at full eligible pool scale.

2.1 What is Spotify Reserved?

Reserved is a Spotify Premium feature that uses in-app behavioural signals - stream frequency, playlist saves, share activity, discography breadth, and concert page engagement - to rank listeners by artist devotion. The top-ranked fans per artist receive a time-limited notification to purchase two tickets at face value before general sale, in partnership with Live Nation (signed as multi-year exclusive launch partner).

Metric	Value	Source
Total MAU	761 million	Q1 2026 Earnings
Premium Subscribers	293 million	Q1 2026 Earnings
Monthly ARPU	\$4.58	Q1 2026 Earnings
All-time ticket revenue driven	\$1.5B+	Investor Day 2026
Ticketing partners	40+	Investor Day 2026
2030 subscriber target	1 billion	Investor Day 2026
Stock move on announcement	+15%	Market data

Table 1: Spotify business metrics at time of Reserved launch

2.2 Research Questions

#	Research Question
Q1	What listening signals most strongly predict superfan status?
Q2	How could Spotify improve the Reserved experience beyond ticket access?
Q3	What structural fairness risks exist in the eligibility model?
Q4	How does Spotify collect and score behavioural signals per user?
Q5	How does the model perform at A-list scale (50M+ monthly listeners)?
Q6	Does a Reserved offer improve 90-day Premium retention?
Q7	What is the revenue uplift for artists, venues, and Spotify?

2.3 Stakeholder Map

Stakeholder	Type	Influence	Primary Question
Spotify Product Team	Internal	High	Does Reserved reduce churn?
Spotify Live Events	Internal	High	How do we score eligibility fairly?
Live Nation	Partner	High	Does Reserved displace scalpers?
Artists & Managers	External	Medium	Will my real fans get tickets?
Premium Subscribers	End User	Medium	Am I eligible? How do I qualify?
Spotify Investors	Financial	Lower	Does Reserved drive the 1B sub target?

3.1 Data Sources

Source	Data Collected	Access	Quality
Python Synthetic Data Generation	Artist profiles, audio features, popularity, top tracks	Free API	High
Simulated Listener Pool	30,000 listener records with 6 engagement signals	Constructed	Controlled
Spotify Investor Relations	Q1 2026 earnings, ARPU, subscriber counts	Public	High
Industry Benchmarks	Concert fill rates, in-venue spend, acceptance rates	Published	Medium

Table 2: Data sources

3.2 Three Scoring Approaches Compared

Three independent scoring methods were built, compared, and one selected based on explainability and statistical validation.

Approach	Method	Validation	Selected?
A: Weighted Linear	6-signal composite with Phase 1 pre-set weights	Rule-based	YES - most explainable
B: RFM Quintile	Recency/Frequency/Monetary quintile scoring	Rule-based	Used for validation
C: Logistic Regression	Data-driven binary classifier on eligibility flag	AUC = 0.87	Used for validation

Model A was selected because explainability is paramount when presenting to Spotify's product team. The LR model (AUC = 0.87) validates that the 6 signals genuinely discriminate superfans - confirming the weighted model's signal selection is sound.

3.3 RFM Framework Applied to Music Streaming

RFM Dimension	Music Equivalent	Signal	Weight
Recency	How recently active with artist	Follow tenure (months)	10%
Frequency	How often they replay tracks	Repeat play rate	30%
Frequency (2nd)	Do they check concert events?	Concert page views	5%
Monetary	Do they save tracks to playlists?	Playlist save rate	25%
Monetary (2nd)	Do they share the music?	Share rate	15%
Monetary (3rd)	How much of catalogue heard?	Discography breadth	15%

Table 3: RFM framework mapped to music streaming signals

3.4 Assumptions Log Summary

ID	Assumption	Impact	Risk if Wrong
A1	Spotify Charts = proportional engagement proxy	Medium	Underweights listeners outside Top 200
A3	Top 5% per artist = superfan threshold (p95)	High	May need genre-level calibration
A4	35-55% Reserved offer acceptance rate	High	Could be higher or lower with real data
A5	Superfan in-venue spend +20-30% vs general	Medium	Varies by genre and venue type
A7	Reserved improves 90-day retention by 5-8%	High	Smaller if fans cancel post-concert
A8	3-8% of Premium listeners qualify per artist	Medium	Varies significantly by artist scale

Full assumptions log in docs/01_business_understanding.md

4.1 Superfan Score Distribution

The composite superfan score was calculated for 30,000 simulated listeners across 30 artists. The distribution is strongly right-skewed - most listeners are casual (score < 30), while only a small elite tail reaches superfan territory (score > 58). This shape validates the p95 threshold as a natural, meaningful cutoff.

Statistic	Value	Interpretation
Mean	~28.4	Typical listener has low engagement intensity
Median	~26.1	Over 50% of listeners score below 27
p90 (top 10%)	~48.0	Engaged but not yet Reserve-eligible
p95 - Reserved threshold	~57.8	Top 5% - eligibility cutoff
p99 (top 1%)	~72.3	True superfan elite
Maximum	~91.4	Near-perfect engagement across all signals

Table 4: Score distribution statistics (30,000-listener pool)

4.2 Signal Weight Analysis

The logistic regression validation model learns signal weights directly from data. Its coefficients closely mirror the Phase 1 design weights - confirming that the a-priori weighting was well-calibrated.

Signal	Design Weight	LR Coefficient	Interpretation
Repeat play rate	30%	Highest (2.84)	Replaying = clearest signal of genuine love for the music
Playlist save rate	25%	High (2.31)	Saving = intent to return; strongest long-term fandom proxy
Share rate	15%	Moderate (1.62)	Sharing is rare - makes it a high-signal indicator when it occurs
Discography breadth	15%	Moderate (1.58)	Catalogue-wide listening signals depth over recency
Follow tenure (months)	10%	Lower (1.12)	Early followers loyal but recency can distort the signal
Concert page views	5%	Lowest (0.87)	Strong intent signal but very low base rate across all users

Key insight: Repeat plays + playlist saves = 55% of the score. These are also the two signals Spotify measures most reliably - no user input, survey, or permission required beyond standard app usage. This makes the model robust and privacy-safe.

4.3 Artist Segmentation - K-Means Clustering

K-Means clustering (k=4, selected by silhouette score) was applied to 30 artists using 9 features spanning audio characteristics, engagement metrics, and scale. Clusters reveal nuances within manual tiers - particularly that some "Mainstream" artists have developing-level engagement intensity.

Cluster	Artists (sample)	Avg Followers	Avg Score	Reserved Verdict
Stadium Headliners	Weeknd, T.Swift, Drake	112M	29.2	High demand; needs fairness mechanism
Mainstream Giants	Bad Bunny, SZA, Doja Cat	42M	31.8	Core Reserved use case
Cult Favourites	Tame Impala, Phoebe Bridgers	6.5M	34.1	Highest engagement intensity
Emerging Voices	Ethel Cain, Gracie Abrams	530K	35.6	Strongest ROI - loyal small fanbase

4.4 A-List Scale Stress Test

The most critical scalability challenge for Reserved is the extreme supply-demand gap at A-list level. The table below shows the competition ratio - eligible fans per available offer - for a representative cross-section of artists.

Artist	Tier	Est. Eligible Fans	Offers/Show	Competition Ratio
Taylor Swift	A-List Stadium	1,725,000	1,625	1,062:1
The Weeknd	A-List Stadium	1,590,000	1,625	978:1
Bad Bunny	Mainstream	742,500	625	1,188:1
SZA	Mainstream	567,000	625	907:1
Tame Impala	Mid-Size	108,000	125	864:1
Arlo Parks	Developing	5,400	20	270:1
Gracie Abrams	Developing	9,450	20	473:1

Table 6: A-list stress test - competition ratio by artist

5.1 Model 1 - Premium Subscriber Retention & ARR Impact

Asimulated A/B test (two-proportion z-test, $p < 0.05$) confirms a statistically significant 6% improvement in 90-day retention for Reserved offer recipients vs a matched control group. The table below shows ARR implications across retention scenarios.

Scenario	Retention Uplift	Fans Retained (Y2+)	Incremental ARR
Bear Case	+ 3%	4,395,000	\$242M / year
Base Case	+ 6%	8,790,000	\$484M / year
Bull Case	+ 10%	14,650,000	\$806M / year

Table 7: ARR sensitivity - full eligible pool (14.65M fans), ARPU \$4.58/mo

Base case: \$484M incremental ARR. Even the bear case (\$242M) represents a compelling return on a feature with near-zero marginal cost to Spotify per user served.

5.2 Model 2 - Tour Revenue Uplift (50-Date Arena Simulation)

Three revenue channels are modelled per show: higher fill rate (Reserved fans almost never no-show), elevated in-venue spend (+\$27/head for superfans), and Spotify's transaction fee on Reserved-window ticket sales.

Revenue Component	Baseline (No Reserved)	With Reserved	Uplift/Show
Ticket revenue	\$2,125,000	\$2,194,375	+\$69,375
In-venue spend	\$987,000	\$1,028,410	+\$41,410
Spotify fee (5% Reserved)	\$0	\$5,513	+\$5,513
Total per show	\$3,112,000	\$3,228,298	+\$116,298
50-date tour total	\$155,600,000	\$161,414,900	+\$5,814,900

Table 8: Tour revenue model - 50 dates, 20,000-seat arena, \$125 avg ticket

5.3 Model 3 - Secondary Market Displacement

By routing tickets directly to verified high-engagement fans, Reserved reduces scalper supply. Pre-Reserved, ~25% of high-demand tickets end up on resale at 2.8x markup. Post-Reserved, this falls to ~5%.

Metric	Without Reserved	With Reserved	Fan Benefit
Tickets scalped/show	5,000 (25%)	1,000 (5%)	4,000 fewer scalped
Fan overpayment/show	\$1,125,000	\$225,000	Fans save \$900K/show
Fan savings, 50-date tour	-	-	\$45,000,000 total saved

Table 9: Secondary market displacement (20,000-seat venue, 2.8x scalper markup)

Six structural fairness risks were identified and rated. Each has a proposed mitigation.

Risk	Severity	Likelihood	Proposed Mitigation
Bot streaming farms	High	Medium	Behavioural consistency checks; account age gate; device fingerprinting
Algorithm opacity	High	Certain	Publish scoring principles; add in-app score feedback feature
Geographic inequity	Medium	Certain	Geographic quota distribution; virtual show Reserved tier
Recency bias	Medium	Likely	Loyalty decay weighting; multi-year engagement history windows
Supply-demand gap (A-list)	High	Certain	Loyalty Rotation System; regional quotas; tiered allocation
Off-platform fandom blind spot	Low	Certain	Acknowledge limitation; future cross-platform data partnerships

Three Proposed Fairness Mechanisms

Loyalty Rotation System

Eligible fans who received Reserved in a previous tour cycle are de-prioritised in the next. Long-term fans who have never received an offer get priority. Best for legacy artists (Adele, Ed Sheeran).

Geographic Quota Distribution

Reserved tickets distributed proportionally across all tour markets before per-city competition. Prevents all offers concentrating in the 3 largest cities.

Score Decay for Sudden Converts

Listeners who begin heavily streaming an artist within 60 days of tour announcement have their score decayed 20%/week. Protects long-term fans from bot spikes and bandwagon listeners.

7

Recommendations

R1 Adopt a transparent two-layer eligibility model

Hard gates (Premium + location + age + notifications on) filter the addressable pool. Within that pool, publish the scoring principles - not exact weights - to build fan trust.

R2 Implement Loyalty Rotation for A-list artists before launch

A 1,000:1 competition ratio is untenable. Rotating Reserved offers across tour cycles ensures all eligible fans get access over time rather than the same fans repeatedly winning.

R3 Extend the Reserved experience beyond ticket purchase

Version 2.0: exclusive pre-show artist audio message, priority venue entry lane, personalised post-concert recap. Each costs Spotify near-zero and dramatically increases perceived value.

R4 Prioritise developing and mid-size artists in Year 1

These artist segments have the highest engagement intensity and smallest supply-demand gap. Starting here maximises fairness perception and builds the data foundation for A-list rollout.

R5 Build score decay before public launch, not after

Bot gaming will accelerate once Reserved becomes widely known. Pre-launch score decay is far cheaper than post-hoc fraud detection and appeal infrastructure.

R6 Track all 6 Phase 1 KPIs from Day 1

Offer acceptance rate, 90-day retention uplift, Reserved-driven Premium upgrades, secondary market displacement, superfan score delta post-concert, geographic coverage rate.

R7 Use Reserved as the primary free-to-Premium upsell hook

"Upgrade for a chance to get Reserved tickets to your favourite artist" is Spotify's most compelling upsell. Deploy this in free-tier discovery flows near tour announcements.

8.1 Summary of Conclusions

Spotify Reserved is a genuine strategic inflection point - not just a product feature but a proof of concept for a new monetisation model: converting Spotify's data advantage into tangible real-world value for fans. This analysis confirms that the underlying signals are sufficient to identify superfans reliably, the retention and revenue uplift are meaningful and statistically significant, and the feature is viable across all artist tiers. The \$484M incremental ARR base case, \$5.8M tour revenue uplift, and \$45M fan scalper savings per major tour make the investment case straightforward.

8.2 Project Limitations

- Alllistener data is simulated - not Spotify's real internal clickstream data.
- Financial models are projections based on published figures and industry benchmarks.
- Analysis covers the US market only. International dynamics will differ significantly.
- Artist sample of 30 is intentionally diverse but not statistically representative of all Spotify artists.

8.3 Suggested Next Steps for Spotify's Data Team

Immediate (0–3 months)

- Baseline all 6 Phase 1 KPIs before the first Reserved offer is sent
- A/B test score decay mechanism on a small artist cohort
- Implement geographic quota logic in the allocation algorithm

Near-term (3–12 months)

- Launch thefirst Reserved experience enhancement (post-concert recap)
- Analyse actual acceptance rates vs model assumption of 45%
- Extend Reserved to 500+ artists across all four tiers

Long-term (12+ months)

- Build Loyalty Rotation System for artists with >100K competing eligible fans
- Explore cross-platform fandom signals as eligibility inputs
- Model Reserved as primary conversion driver toward the 1 billion subscriber target

GitHub Repository

<https://github.com/chiragmendon16/spotify-reserved-analysis>

Tableau Dashboard

[public.tableau.com/\[yourname\]/SpotifyReserved](https://public.tableau.com/[yourname]/SpotifyReserved)

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